

JOINS DEEP DIVE

Comprehensive guide to INNER, LEFT, RIGHT, FULL, and SELF JOIN operations with real-world scenarios.

INNER JOIN
Exact Matches

LEFT JOIN
All Left Rows

RIGHT JOIN
All Right Rows

FULL JOIN
All Union Rows

SELF JOIN
Same-Table Link

1. Food Delivery App (Swiggy / Zomato)

INNER JOIN

Fetch order_id, customer_name, and product_name by joining Orders, Customers, and Products tables.

```
SELECT
  o.order_id,
  c.customer_name,
  p.product_name
FROM Orders o
INNER JOIN Customers c ON o.customer_id = c.customer_id
INNER JOIN Products p ON o.product_id = p.product_id;
```

Execution Result Set:

order_id	customer_name	product_name
1001	Ananya	Paneer Butter Masala
1002	Rohan	Chicken Biryani

2. E-Commerce User Orders (Flipkart Style)

LEFT JOIN

List all users from the Users table along with their latest order date, keeping users who have never placed an order.

```
SELECT
  u.username,
  MAX(o.order_date) AS last_order_date
FROM Users u
LEFT JOIN Orders o ON u.user_id = o.user_id
GROUP BY u.user_id, u.username;
```

Execution Result Set:

username	last_order_date
rahul_99	2026-08-15
priya_s	2026-09-01
amit_v	NULL

3. Fashion Catalog Purchases (Myntra Style)

[RIGHT JOIN](#)

Display all products and the name of the user who last purchased each item, including items that have never been bought.

```
SELECT
    p.product_name,
    u.username AS last_bought_by
FROM Orders o
INNER JOIN Users u ON o.user_id = u.user_id
RIGHT JOIN Products p ON o.product_id = p.product_id;
```

Execution Result Set:

product_name	last_bought_by
Denim Jacket	sneha_m
Running Shoes	vikram_r
Leather Wallet	NULL

4. Payment & Refund Reconciliation (Paytm Style)

[FULL JOIN](#)

Combine payment transactions and refund records to output transaction_id, amount, and the record type classification.

```
SELECT
    COALESCE(p.transaction_id, r.transaction_id) AS transaction_id,
    COALESCE(p.payment_amount, r.refund_amount) AS amount,
    CASE
        WHEN p.transaction_id IS NOT NULL AND r.transaction_id IS NOT NULL
            THEN 'Both (Payment & Refund)'
        WHEN p.transaction_id IS NOT NULL
            THEN 'Payment Only'
        WHEN r.transaction_id IS NOT NULL
            THEN 'Refund Only'
        END AS transaction_type
FROM Payments p
FULL OUTER JOIN Refunds r ON p.transaction_id = r.transaction_id;
```

Execution Result Set:

transaction_id	amount	transaction_type
TXN101	₹500.00	Both (Payment & Refund)
TXN102	₹1200.00	Payment Only
TXN103	₹350.00	Refund Only

5. Same Genre Movies Pairings (BookMyShow Style)

[SELF JOIN](#)

Find all movie pairs that share the same genre without matching a movie to itself or generating duplicate inverse pairs.

```
SELECT
    m1.movie_title AS movie_1,
    m2.movie_title AS movie_2,
    m1.genre
```